April/May Exam Session 2024
Duration: 1 hour 30 minutes
Additional time: 30 minutes
Timed exam fixed start time

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

TEXT AS DATA FOR MSC/MSCI
COMPSCI 5096/5106

(Answer all 3 questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1. Question on Tokenisation. (Total marks: 20)

(a) Why are stopwords normally removed when encoding text in a bag-of-words representation? Why are stopwords normally **not** removed when encoding text with a transformer network? Provide an illustrative example.

[4]

> **Example Answer:**
>
> Consider the phrase "The cat sat on the mat"
>
> In bags-of-words, the attachment of the stopwords is lost. E.g., it would be unclear if the cat is on the mat or the mat is on the cat.
>
> Transformers consider the sequence of the words in context. The model can represent the cat being on the mat (not the mat on the cat) since the word order is considered.

(b) One of your colleagues states that the `[UNK]` token is not necessary for subword tokenisers like BPE (Byte Pair Encoding). Considering what you know about Unicode encoding, are they correct? Why or why not?

[4]

> **Example Answer:**
>
> The colleague is incorrect. UNK is used when a character is found that was not seen in the original text. Unicode can encode more characters than are likely to appear in training data. Further, new Unicode characters are frequently added which will not appear in older training data.

(c) Perform sub-word tokenisation over the following two tokens using the following BPE rules. Assume all characters `[m,i,s,p,e]` exist in the vocabulary. Show all your steps.

[6]

BPE Rules:

```
i, s -> is
p, i -> pi
i, e -> ie
p, ie -> pie
p, pi -> ppi
si, ppi -> sippi
```

Tokens to encode:

```
mississippi pie
```

> **Example Answer:**
>
> **mississippi**

```
  m i s s i s s i p p i
Apply Rule (i, s -> is)
  m is s is s i p p i
Apply Rule (p, i -> pi)
  m is s is s i p pi
Rule (i, e -> ie) does not apply
Rule (p, ie -> pie) does not apply
Apply Rule (p, pi -> ppi)
  m is s is s i ppi
Rule (si, ppi -> sippi) does not apply
(done)
```

pie

```
  p i e
Rule (i, s -> is) does not apply
Apply Rule (p, i -> pi)
  pi e
Rule (i, e -> ie) does not apply
Rule (p, ie -> pie) does not apply
(The remainder of the rules do not apply either)
(done)
```

(d) What would be the effect of setting BPE's target vocabulary size too low (e.g., 100) when training? What about too high (e.g., 100,000,000)?

[6]

Example Answer:

Too Low: If BPE's vocabulary size is too low, words will be split into too many subwords. At 100, BPE would probably only be able to represent individual characters, which carry essentially no meaning. It will be difficult for text processing systems to operate over units with such little meaning.

Too High: When set too high, words observed in the training data will not be split up into any subwords. 100,000,000 is higher than the total number of words in many corpora, so each word will get its own token. This approach defeats the purpose of subword tokenization, because it will be difficult for text processing systems to learn good representations of so many tokens.

2. Question on Language Modelling with Character N-grams. (Total marks: 20)

Consider the text below that is 20 characters long. We will build n-gram language models using this text and then generate new text to follow it.

BCBBCBBABCCCBBCCBCCB

(a) Generate the next three tokens for the 20 character text above. Use a character **unigram** model and a **greedy approach**

[2]

Example Answer:
$P(A) = \frac{1}{20}$
$P(B) = \frac{10}{20} = \frac{1}{2}$
$P(C) = \frac{9}{20}$
The next three tokens would be "B B B".

(b) Generate the next three tokens for the 20 character text above. Use a character **trigram** model and a **greedy approach**. You do not need to calculate the full trigram model.

[6]

Example Answer:
$P(B|CB) = \frac{3}{4}$
$P(C|CB) = \frac{1}{4}$
First new token is B.

$P(A|BB) = \frac{1}{3}$
$P(C|BB) = \frac{2}{3}$
Second new token is C.

$P(B|BC) = \frac{2}{5}$
$P(C|BC) = \frac{3}{5}$
Third new token is C.

The next three tokens would be "B C C"

(c) Generate the next **two** tokens for the 20 character text above. Use a character **bigram** model and **beam search** with a beam width of 2. Show the calculations and active beams at each step and then the final output with the highest probability.

[8]

Example Answer:
First token:
$P(A|B) = \frac{1}{9}$
$P(B|B) = \frac{3}{9} = \frac{1}{3}$
$P(C|B) = \frac{5}{9}$

Beams:

- A ($P = \frac{1}{9}$)
- B ($P = \frac{3}{9} = \frac{1}{3}$)
- C ($P = \frac{5}{9}$)

Keep Beams: C and B

Second token:
$P(B|C) = \frac{5}{9}$
$P(C|C) = \frac{4}{9}$

Beams:

- CB ($P = \frac{5}{9} \times \frac{5}{9} = \frac{25}{81}$)
- CC ($P = \frac{5}{9} \times \frac{4}{9} = \frac{20}{81}$)
- BA ($P = \frac{3}{9} \times \frac{1}{9} = \frac{3}{81} = \frac{1}{27}$)
- BB ($P = \frac{3}{9} \times \frac{3}{9} = \frac{9}{81} = \frac{1}{9}$)
- BC ($P = \frac{3}{9} \times \frac{5}{9} = \frac{15}{81} = \frac{5}{27}$)

Beam with highest probability is CB ($P = \frac{25}{81}$)

(d) Propose a two character text that a bigram model would be unable to generate (following the 20 character text above). It should use the same alphabet as before. Explain why a model is unable to generate it. Briefly propose a method for adjusting the probabilities and the text generation method which means that the language model may generate it.

[4]

Example Answer:
One string (of several) that a bigram model would be unable to generate is ”AA”. It

cannot generate that as $P(A|A) = 0$.

Smoothing could be applied to the probabilities so that they are non-zero. The probability of the string would still be small so a sampling-based text generation approach could be used so that there is some possibility of the text being generated.

CONTINUED OVERLEAF

3. Question on Indirect Answer Classification. (Total marks: 20)

Recall the task of Indirect Answer Classification (IAC) from the paper *"Id rather just go to bed": Understanding Indirect Answers* (Louis et al., EMNLP 2020), which you read for Coursework 2. (The paper is available on Moodle if you need to refer back to it.)

(a) Consider a situation where you try a simple prompt on a large language model to perform IAC on *Circa*, and you obtain a higher effectiveness on the test set than what was reported by Louis et al. What do you need to consider before asserting that the effectiveness is "zero-shot" (obtained without training the model without any IAC data)?

[2]

> **Example Answer:** There may be data contamination, wherein the IAC data was part of the data used for training the large language model.

(b) A colleague suggests that *Circa*'s Fleiss kappa of 0.61 means that the dataset's labels are unreliable. They suggest using a large language model to provide the labels instead. Provide three counter-arguments.

[6]

> **Example Answer:**
>
> - 0.61 is actually a reasonably high agreement score. Typically seen as indicating "Substantial agreement".
> - In the "relaxed" setting, the agreement is much higher at 0.76, so annotators typically agree on the polarity but disagree on the degree.
> - Disagreement as identified by the Fleiss kappa score could indicate that the task was not sufficiently defined. Using a large language model would not address an imprecise task definition.

(c) Recall that the *unmatched* evaluation setting aims to evaluate a model's performance on unseen scenarios. Using what you learned about text similarity, why might their approach be insufficient? Propose a technique that could better evaluate performance on unseen scenarios.

[6]

> **Example Answer:**
> Although the exact same question cannot appear in multiple sets, very similar questions can, which may defeat the purpose of the unmatched setting.
> **Greedy Algorithm:** Randomly shuffle the questions. Assign one question to the train, development, and test sets. For the remaining questions, assign it to the set with the

lowest maximum similarity score (e.g., cosine of TF-IDF vectors) to any other question. Although such a greedy solution will not maximise the distance between each set, it helps ensure some distance is achieved, and it is relatively efficient to run.

(d) The paper tests models that use multiple stages of fine-tuning on different datasets. What is the motivation for this approach? Why is effectiveness degraded when fine-tuning on some datasets?

[6]

Example Answer:

Applying multiple stages of fine-tuning can allow a model to transfer knowledge from one task to another. This is similar to the idea of language model pre-training itself (transferring knowledge from one task to another).

Some datasets (e.g., MNLI) improve effectiveness since the knowledge transfer benefits learning the IAC task. Meanwhile, others (e.g., BOOLQ) have their effectiveness degraded because the knowledge is detrimental to learning the IAC task. In the case of BOOLQ, the answers are direct instead of indirect, which is not helpful for IAC.